# MATH 231 – Linear Algebra I

Homework 4 ▪ Transformations, Determinants, Systems and Least Squares

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**Due:** Sunday, November 29, 2026, 11:59 PM.

| Matrix Unit | Sections | Topics |
|---|---|---|
| Part 3 | §15–§21 | a matrix from what it does; scaling, translation, shear; orthogonal matrices |
| Part 4 | §22–§25 | the $2 \times 2$ determinant and area; the rotation matrix; rotations of $\mathbb{R}^3$ |
| Part 5 | §26–§29 | projection matrices; reflections; the Householder reflector; the direct sum |
| Part 6 | §30–§34 | the null space; rank–nullity; rank($\mathbf{A}^\mathsf{T}\mathbf{A}$); the $2 \times 2$ inverse |
| Part 7 | §35–§39 | cofactors; the Laplace expansion; principal minors; the laws D1–D10 |
| Part 8 | §40–§45 | row operations; Gaussian and Gauss–Jordan elimination; consistency |
| Part 9 | §46–§49 | least squares; the normal equations; fitting a line |
| Part 10 | §50–§54 | Cramer's rule; the inverse; the adjugate; the pseudo-inverse |
| Lecture | — | linear maps and operators; $C^\infty(\mathbb{R})$; affine maps and their matrix $[\,\mathbf{A} : \mathbf{b}\,]$ |

**Grading.** Every problem is worth the same number of points.

**Section numbers and notation.** A bare section number such as §28.4 refers to the matrix unit, whose numbering runs straight through Parts 1–11. $\mathbf{e}_i$ is the standard basis vector with a 1 in slot $i$. For a matrix $\mathbf{A}$, col[$\mathbf{A}$], row[$\mathbf{A}$] and null[$\mathbf{A}$] are its column space, row space and null space.

**Rows and columns.** A vector $\mathbf{x} \in \mathbb{R}^n$ is a *column*, an $n \times 1$ matrix. A *row* vector is a $1 \times n$ matrix, an element of $\mathbb{R}^{1\times n}$. A bracket written across the line is a row, so a column written across the line carries a transpose: $\mathbf{x} = [\,x_1, \ldots, x_n\,]^\mathsf{T} \in \mathbb{R}^n$. For example, $[\,3, 1\,] \in \mathbb{R}^{1\times 2}$ is a row, while $[\,3, 1\,]^\mathsf{T} = \left[\begin{smallmatrix}3\\1\end{smallmatrix}\right] \in \mathbb{R}^2$ is a column, and $\mathbf{u}^\mathsf{T}\mathbf{v}$ is a number while $\mathbf{u}\mathbf{v}^\mathsf{T}$ is a matrix. Keep the two apart in your answers too: a column written across the line needs its $^\mathsf{T}$.

**On the proof problems** (A.1(e), A.3(a), (b), A.4(a), (b), (d)–(f), B.2(a)–(c), B.3(b), C.3(b), (c), D.2(a)–(c), E.3(d), (e), and F.2(a)). None of them requires a clever idea. Write down what you are assuming, justify each step by naming the definition, law or theorem that licenses it, and end with a sentence saying what you have shown. A chain of equalities with no words is not a proof; neither is a paragraph of words with no computation.

**Collaboration** is encouraged — work at a board together, argue about the problems. What you submit must be written by you, in your own words.

### What to submit

Push your written work for Parts A–F to your own MATH 231 GitHub repository, as a single PDF named `hw4.pdf` in a folder named `homework/hw4/`. Handwritten and scanned or photographed is completely fine, as long as it is legible and the pages are in order.

**PDF is the only accepted format.** A `.docx`, a `.jpg`, a `.png`, a `.heic`, or a link to a file hosted somewhere else will not be graded.

**Type Parts A–F up instead and you earn +5 points of extra credit on this assignment.** Any typesetting system counts — LaTeX, Word's equation editor, or anything else that produces readable mathematics — and the extra credit is for the whole of Parts A–F, not for typing up some of it. Sketches may be drawn by hand and scanned into the PDF.

**Label every answer**

Label each answer with the number of the problem it answers — `A.3(b)`, `E.2(d)` — and put your answers **in the same order as they appear in this assignment**. A grader who cannot tell which question a piece of work belongs to cannot award it credit. You do not have to copy the question out: write the label, then your work.

## Part A. Matrices that move the plane; linear and affine maps

*Matrix Unit, §15–§21 and §24–§25.* Identify the point $(x, y)$ with the column $[x, y]^{\mathsf{T}}$. A matrix $\mathbf{M} \in \mathbb{R}^{2\times 2}$ moves each point $\mathbf{p}$ to $\mathbf{Mp}$, and its columns are the points to which it moves $\mathbf{e}_1$ and $\mathbf{e}_2$ (the recipe of §15.1). To move a polygon, list its corners as the columns of a matrix $\mathbf{V} \in \mathbb{R}^{2\times k}$, compute the single product $\mathbf{MV}$, and join the moved corners in the same order. Doing $\mathbf{B}$ first and then $\mathbf{A}$ is the single matrix $\mathbf{AB}$ (§15.4). A square matrix $\mathbf{Q}$ is **orthogonal** if $\mathbf{Q}^{\mathsf{T}}\mathbf{Q} = \mathbf{Q}\mathbf{Q}^{\mathsf{T}} = \mathbf{I}$ (§20.6), and the $2\times 2$ orthogonal matrices are exactly the rotations $\left[\begin{smallmatrix} a & -c \\ c & a \end{smallmatrix}\right]$ and the reflections $\left[\begin{smallmatrix} a & c \\ c & -a \end{smallmatrix}\right]$ with $a^2 + c^2 = 1$ (§21.2). In the angle form of §24.2 and §24.7,

$$\mathbf{R}_\theta = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}, \qquad \mathbf{F}_\theta = \begin{bmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{bmatrix}:$$

$\mathbf{R}_\theta$ rotates counterclockwise by $\theta$, and $\mathbf{F}_\theta$ reflects across the line through the origin at angle $\theta/2$. Two shapes are used below: the unit square $S$ and the standard triangle $\Delta$, with corner matrices $\mathbf{V}_S = \left[\begin{smallmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{smallmatrix}\right]$ and $\mathbf{V}_\Delta = \left[\begin{smallmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{smallmatrix}\right]$. Draw every sketch on graph paper, or on hand-drawn axes with the grid marked, with the original shape and the moved shape on the same axes.

**Problem A.1** (A gallery of motions)**.**
Here are six matrices:

$$\mathbf{M}_1 = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}, \quad \mathbf{M}_2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \mathbf{M}_3 = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix},$$

$$\mathbf{M}_4 = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{M}_5 = \frac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \quad \mathbf{M}_6 = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}.$$

**(a)** For each $\mathbf{M}_i$, compute $\mathbf{M}_i\mathbf{V}_S$, sketch $S$ and its image, and say what $\mathbf{M}_i$ does to the square: a stretch (by how much in each direction), a reflection (across which line), a rotation (by which angle, and which way), a shear (which way), or a collapse onto a line (which line).

**(b)** For each $\mathbf{M}_i$, read the area of the image of $S$ off your sketch, and compute $\det \mathbf{M}_i$. Check that the area is $|\det \mathbf{M}_i|$ (§23.4). Which of the six reverse orientation?

**(c)** Which of the six are orthogonal? Write each one that is as $\mathbf{R}_\theta$ or $\mathbf{F}_\theta$ for a specific angle $\theta$.

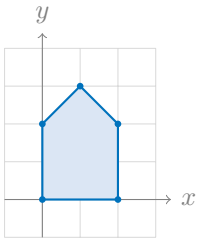
**(d)** Find the matrix that moves $\Delta$ onto the triangle with corners $(0, 0)$, $(3, 1)$, $(1, 2)$, and the one that moves $\Delta$ onto the triangle with corners $(0, 0)$, $(-1, 2)$, $(4, 0)$. Use a determinant to find the area of each of the two triangles.

**(e)** Prove that no $2\times 2$ matrix moves $\Delta$ onto the triangle with corners $(1, 1)$, $(2, 1)$, $(1, 2)$. Then write a $3\times 3$ matrix, in homogeneous coordinates (§17.3), that does.

**Problem A.2** (One motion after another)**.**
The house below has corners $(0,0)$, $(2,0)$, $(2,2)$, $(1,3)$, $(0,2)$, taken in that order. Let

$$\mathbf{V}_H = \begin{bmatrix} 0 & 2 & 2 & 1 & 0 \\ 0 & 0 & 2 & 3 & 2 \end{bmatrix}, \qquad \mathbf{R} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \qquad \mathbf{F} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix},$$

so that $\mathbf{R}$ is the quarter turn counterclockwise and $\mathbf{F}$ is the reflection across the $x$-axis.




**(a)** Compute $\mathbf{R}\mathbf{V}_H$ and sketch the turned house.

**(b)** Compute $\mathbf{F}(\mathbf{R}\mathbf{V}_H)$, which turns the house and then reflects it, and $\mathbf{R}(\mathbf{F}\mathbf{V}_H)$, which reflects it and then turns it. Sketch both. Do they agree?

**(c)** Compute the single matrices $\mathbf{FR}$ and $\mathbf{RF}$, and check that $(\mathbf{FR})\mathbf{V}_H = \mathbf{F}(\mathbf{R}\mathbf{V}_H)$. Each of $\mathbf{FR}$ and $\mathbf{RF}$ is a reflection: write each as $\mathbf{F}_\theta$, and name the line it reflects across.

**(d)** Since $\mathbf{R} = \mathbf{R}_{\pi/2}$, use §24.6 to write $\mathbf{R}^2$, $\mathbf{R}^3$ and $\mathbf{R}^4$ as rotations $\mathbf{R}_\theta$ without multiplying. Then check your answer for $\mathbf{R}^2$ by multiplying.

**(e)** Write the $3 \times 3$ matrix, in homogeneous coordinates (§17.3–§17.4), that first turns the house by $\mathbf{R}$ and then translates it by $[\,3, 1\,]^{\mathsf{T}}$. Apply it to all five corners with one matrix product, and sketch the result.

**Problem A.3** (Orthogonal matrices)**.**

**(a)** Let $\mathbf{Q}_1$ and $\mathbf{Q}_2$ be orthogonal $n \times n$ matrices. Prove that $\mathbf{Q}_1\mathbf{Q}_2$ is orthogonal, naming the laws of §8.4 you use.

**(b)** Using the angle-addition formulas $\cos(\alpha - \beta) = \cos\alpha\cos\beta + \sin\alpha\sin\beta$ and $\sin(\alpha - \beta) = \sin\alpha\cos\beta - \cos\alpha\sin\beta$, prove that $\mathbf{F}_\alpha\mathbf{F}_\beta = \mathbf{R}_{\alpha-\beta}$: two reflections make a rotation. Check it for $\alpha = \pi/2$ and $\beta = 0$ by multiplying.

**(c)** Find every $2 \times 2$ orthogonal matrix whose first column is $[\,\frac{5}{13}, \frac{12}{13}\,]^{\mathsf{T}}$. For the rotation, give $\cos\theta$ and $\sin\theta$. For the reflection, find a nonzero vector on its mirror line and check that the matrix leaves it where it is.

**(d)** Let $\mathbf{M} = \mathbf{R}_x(\frac{\pi}{2})\,\mathbf{R}_z(\frac{\pi}{2})$, with

$$\mathbf{R}_x(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}, \qquad \mathbf{R}_z(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

the rotations of §25.1–§25.2. Compute $\mathbf{M}$ and $\mathbf{M}^{\mathsf{T}}\mathbf{M}$. Find every vector $\mathbf{v} \in \mathbb{R}^3$ with $\mathbf{M}\mathbf{v} = \mathbf{v}$, and compute $\mathbf{M}^3$. ($\mathbf{M}$ is the rotation of space by $2\pi/3$ about the line you found.)

**Problem A.4** (Linear and affine maps)**.**
*From the lecture on linear and affine maps.* Functions are added and scaled point by point: $(f+g)(x) = f(x)+g(x)$ and $(\alpha f)(x) = \alpha f(x)$. $C^\infty(\mathbb{R})$ is the set of functions $f : \mathbb{R} \to \mathbb{R}$ that have derivatives of every order, $f', f'', f''', \ldots$. A map $T$ whose inputs are vectors of $\mathbb{R}^n$, or functions in $C^\infty(\mathbb{R})$, is **linear** if $T(u+v) = T(u)+T(v)$ and $T(\alpha u) = \alpha\, T(u)$ for all inputs $u, v$ and every number $\alpha$. For $\mathbf{A} \in \mathbb{R}^{2\times 2}$ and $\mathbf{b} \in \mathbb{R}^2$, the **affine map** $T(\mathbf{p}) = \mathbf{A}\mathbf{p} + \mathbf{b}$ applies $\mathbf{A}$ to the point $\mathbf{p}$ and then shifts it by $\mathbf{b}$. Its matrix is the $2 \times 3$ matrix $[\,\mathbf{A} : \mathbf{b}\,]$, whose first two columns are those of $\mathbf{A}$ and whose third column is $\mathbf{b}$:

$$[\,\mathbf{A} : \mathbf{b}\,] \begin{bmatrix} x \\ y \\ t \end{bmatrix} = \mathbf{A} \begin{bmatrix} x \\ y \end{bmatrix} + t\,\mathbf{b}, \qquad \text{so} \qquad T\big([\,x, y\,]^{\mathsf{T}}\big) = [\,\mathbf{A} : \mathbf{b}\,]\,[\,x, y, 1\,]^{\mathsf{T}}.$$

**(a)** Let $L : C^\infty(\mathbb{R}) \to C^\infty(\mathbb{R})$ be $L(f) = f'' - 3f' + 2f$. Prove that $L$ is linear, naming the rules of differentiation you use. Compute $L(e^x)$, $L(e^{2x})$ and $L(x)$, and then use linearity alone, with no more differentiation, to find $L(5e^x - 2e^{2x} + 4x)$.

**(b)** Prove that every linear map $T$ sends $\mathbf{0}$ to $\mathbf{0}$. Then prove that none of these three maps is linear, by finding specific inputs at which one of the two properties fails: $g : \mathbb{R} \to \mathbb{R}$, $g(x) = 3x + 2$; $h : \mathbb{R}^2 \to \mathbb{R}$, $h\big([\,x, y\,]^{\mathsf{T}}\big) = xy$; and $N : C^\infty(\mathbb{R}) \to C^\infty(\mathbb{R})$, $N(f) = f\,f'$. ($h$ and $N$ do send $\mathbf{0}$ to $\mathbf{0}$, so for them that test is not enough.)

**(c)** Let $\mathbf{A} = \left[\begin{smallmatrix} 1 & -1 \\ 1 & 1 \end{smallmatrix}\right]$, $\mathbf{b} = [\,2, -1\,]^{\mathsf{T}}$ and $T(\mathbf{p}) = \mathbf{A}\mathbf{p} + \mathbf{b}$. Write $[\,\mathbf{A} : \mathbf{b}\,]$. Move the unit square $S$ with one product: append a row of 1s to $\mathbf{V}_S$, multiply by $[\,\mathbf{A} : \mathbf{b}\,]$ on the left, and sketch $S$ and $T(S)$. Compare the area of $T(S)$ with $|\det \mathbf{A}|$.

**(d)** Prove that an affine map $T(\mathbf{p}) = \mathbf{A}\mathbf{p} + \mathbf{b}$ is linear if and only if $\mathbf{b} = \mathbf{0}$. Then prove that the map $\mathbb{R}^3 \to \mathbb{R}^2$, $\mathbf{u} \mapsto [\,\mathbf{A} : \mathbf{b}\,]\,\mathbf{u}$, is linear for every $\mathbf{A}$ and $\mathbf{b}$, naming the laws of §8.4 you use. (So $T$ is a linear map one dimension up, used only on the inputs with $t = 1$; compare the remark in §17.3.)

**(e)** Let $S(\mathbf{p}) = \mathbf{C}\mathbf{p} + \mathbf{d}$ be a second affine map. Prove that doing $T$ and then $S$ is the affine map with matrix $[\,\mathbf{C}\mathbf{A} : \mathbf{C}\mathbf{b} + \mathbf{d}\,]$. The product $[\,\mathbf{C} : \mathbf{d}\,][\,\mathbf{A} : \mathbf{b}\,]$ of two $2 \times 3$ matrices is not defined; the $3 \times 3$ matrices of §17.4 fix this. For the $T$ of (c) and $\mathbf{C} = \left[\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right]$, $\mathbf{d} = [\,0, 1\,]^{\mathsf{T}}$, check that $\left[\begin{smallmatrix} \mathbf{C} & \mathbf{d} \\ \mathbf{0}^{\mathsf{T}} & 1 \end{smallmatrix}\right]\left[\begin{smallmatrix} \mathbf{A} & \mathbf{b} \\ \mathbf{0}^{\mathsf{T}} & 1 \end{smallmatrix}\right]$ has $[\,\mathbf{C}\mathbf{A} : \mathbf{C}\mathbf{b} + \mathbf{d}\,]$ as its top two rows.

**(f)** For the $T$ of (c), find the point $\mathbf{c}$ with $T(\mathbf{c}) = \mathbf{c}$, and prove that $T(\mathbf{p}) = \mathbf{c} + \mathbf{A}(\mathbf{p} - \mathbf{c})$ for every $\mathbf{p} \in \mathbb{R}^2$. Check that $\mathbf{A} = \sqrt{2}\,\mathbf{R}_{\pi/4}$: $T$ turns the plane by $\pi/4$ about $\mathbf{c}$ and stretches every distance from $\mathbf{c}$ by $\sqrt{2}$.

## Part B. Projections, reflections, and the direct sum

*Matrix Unit, §26–§29.* For a nonzero $\mathbf{u} \in \mathbb{R}^n$, the projection matrix onto the line $\operatorname{span}\{\mathbf{u}\}$ is $\mathbf{P}_{\mathbf{u}} = \mathbf{u}\mathbf{u}^{\mathsf{T}}/\mathbf{u}^{\mathsf{T}}\mathbf{u}$, and $\mathbf{P}_{\mathbf{u}}\mathbf{v} = \operatorname{proj}_{\mathbf{u}}(\mathbf{v}) = \frac{\mathbf{u}^{\mathsf{T}}\mathbf{v}}{\mathbf{u}^{\mathsf{T}}\mathbf{u}}\,\mathbf{u}$ (§26.2). A square matrix $\mathbf{P}$ is a **projection matrix** if $\mathbf{P}^2 = \mathbf{P}$ (§27.2), and an **orthogonal projection matrix** if also $\mathbf{P}^{\mathsf{T}} = \mathbf{P}$ (§27.4). By the theorem of §27.4, for an orthogonal projection matrix the leftover $\mathbf{v} - \mathbf{P}\mathbf{v}$ is orthogonal to every vector of $\operatorname{col}[\mathbf{P}]$. The reflection across $\operatorname{span}\{\mathbf{u}\}$ is $2\mathbf{P}_{\mathbf{u}} - \mathbf{I}$ (§28.4). For $\mathbf{u} \in \mathbb{R}^2$ not on the negative $x$-axis, the **Householder reflector** is

$$\mathbf{H} = 2\mathbf{v}\mathbf{v}^{\mathsf{T}} - \mathbf{I}_2, \qquad \mathbf{v} = \frac{\mathbf{m}}{\|\mathbf{m}\|}, \qquad \mathbf{m} = \tfrac{1}{2}\big(\mathbf{u} + \|\mathbf{u}\|\,\mathbf{e}_1\big),$$

and $\mathbf{Hu} = \|\mathbf{u}\|\,\mathbf{e}_1$ (§28.5–§28.6). If $U$ and $W$ are subspaces of $\mathbb{R}^n$, then $\mathbb{R}^n = U \oplus W$, the **direct sum**, means that every vector of $\mathbb{R}^n$ can be written in exactly one way as $\mathbf{u} + \mathbf{w}$ with $\mathbf{u} \in U$ and $\mathbf{w} \in W$. By the theorem of §29.3, this holds if and only if every vector is such a sum and $U \cap W = \{\mathbf{0}\}$.

**Problem B.1** (Building projection matrices)**.**

**(a)** Compute $\mathbf{P_u}$, with exact fractions, for $\mathbf{u} = [\,1, -2\,]^\mathsf{T}$, $[\,2, -1, 2\,]^\mathsf{T}$ and $[\,1, 2, 2\,]^\mathsf{T}$. For each, check that $\mathbf{P_u}$ is symmetric and that $\mathbf{P_u u} = \mathbf{u}$, and give its rank.

**(b)** With $\mathbf{u} = [\,1, 2, 2\,]^\mathsf{T}$ and $\mathbf{v} = [\,3, 0, 3\,]^\mathsf{T}$, compute $\mathbf{P_u v}$ and check it against the formula for $\mathrm{proj}_{\mathbf{u}}(\mathbf{v})$. Then compute $(\mathbf{I} - \mathbf{P_u})\mathbf{v}$ and check that it is orthogonal to $\mathbf{u}$.

**(c)** Let $\mathbf{u} = [\,\cos\theta, \sin\theta\,]^\mathsf{T}$. Show that $\mathbf{P_u} = \begin{bmatrix} \cos^2\theta & \cos\theta\sin\theta \\ \cos\theta\sin\theta & \sin^2\theta \end{bmatrix}$, and that $2\mathbf{P_u} - \mathbf{I} = \mathbf{F}_{2\theta}$, the reflection across the line at angle $\theta$. Use $\cos 2\theta = 2\cos^2\theta - 1 = 1 - 2\sin^2\theta$ and $\sin 2\theta = 2\sin\theta\cos\theta$.

**(d)** Which matrix of Problem A.1 is $\mathbf{P_u}$ for $\mathbf{u} = [\,1, 1\,]^\mathsf{T}$? Which is $2\mathbf{P_u} - \mathbf{I}$ for the same $\mathbf{u}$?

**(e)** Let $\mathbf{1} = [\,1, 1, 1, 1\,]^\mathsf{T} \in \mathbb{R}^4$. Write out $\mathbf{P_1}$, and compute $\mathbf{P_1 x}$ and $(\mathbf{I} - \mathbf{P_1})\mathbf{x}$ for $\mathbf{x} = [\,4, 7, 1, 8\,]^\mathsf{T}$. Then show that for every $\mathbf{x} \in \mathbb{R}^4$, every entry of $\mathbf{P_1 x}$ equals the **mean** $\frac{1}{4}(x_1 + x_2 + x_3 + x_4)$ of the entries of $\mathbf{x}$.

**Problem B.2** (Projections that are not orthogonal)**.**
In (a)–(c), $\mathbf{P}$ is an $n \times n$ matrix with $\mathbf{P}^2 = \mathbf{P}$.

**(a)** Prove that $\mathbf{P}^\mathsf{T}$ is also a projection matrix. Name the law of §8.4 you use.

**(b)** Prove that a vector $\mathbf{v} \in \mathbb{R}^n$ lies in $\mathrm{col}[\mathbf{P}]$ if and only if $\mathbf{Pv} = \mathbf{v}$.

**(c)** Suppose also that $\mathbf{P}^\mathsf{T} = \mathbf{P}$. Prove that $\|\mathbf{Pv}\| \le \|\mathbf{v}\|$ for every $\mathbf{v} \in \mathbb{R}^n$.

**Hint.** Write $\mathbf{v} = \mathbf{Pv} + (\mathbf{v} - \mathbf{Pv})$, and use the theorem of §27.4 and the Pythagorean identity.

**(d)** Let $\mathbf{P} = \left[\begin{smallmatrix} 1 & 2 \\ 0 & 0 \end{smallmatrix}\right]$. Check that $\mathbf{P}^2 = \mathbf{P}$ and that $\mathbf{P}$ is not symmetric. Describe $\mathrm{col}[\mathbf{P}]$, and the set of vectors that $\mathbf{P}$ sends to $\mathbf{0}$, as lines. Compare $\|\mathbf{Pe}_2\|$ with $\|\mathbf{e}_2\|$.

**(e)** With the same $\mathbf{P}$, split $\mathbf{x} = [\,3, 1\,]^\mathsf{T}$ as $\mathbf{Px} + (\mathbf{I} - \mathbf{P})\mathbf{x}$ (§29.5). Are the two pieces orthogonal? Is $\|\mathbf{x}\|^2 = \|\mathbf{Px}\|^2 + \|(\mathbf{I} - \mathbf{P})\mathbf{x}\|^2$?

**Problem B.3** (The Householder reflector, and direct sums)**.**

**(a)** Let $\mathbf{u} = [\,8, 15\,]^\mathsf{T}$. Compute $\mathbf{m}$, $\mathbf{v}$ and $\mathbf{H} = 2\mathbf{v}\mathbf{v}^\mathsf{T} - \mathbf{I}$ with exact entries. Check that $\mathbf{Hu} = [\,17, 0\,]^\mathsf{T}$, that the first column of $\mathbf{H}$ is $\mathbf{u}/\|\mathbf{u}\|$, and that $\mathbf{H}^2 = \mathbf{I}$.

**(b)** Let $\mathbf{u} = r\,[\,\cos\theta, \sin\theta\,]^\mathsf{T}$ with $r > 0$ and $-\pi < \theta < \pi$. Prove that the Householder reflector of $\mathbf{u}$ is $\mathbf{F}_\theta$, so its mirror is the line at angle $\theta/2$, halfway between $\mathbf{u}$ and the positive $x$-axis.

**Hint.** The half-angle identities $1 + \cos\theta = 2\cos^2\frac{\theta}{2}$ and $\sin\theta = 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}$ give $\mathbf{m} = r\cos\frac{\theta}{2}\,\mathbf{w}$, where $\mathbf{w} = [\,\cos\frac{\theta}{2}, \sin\frac{\theta}{2}\,]^\mathsf{T}$. Then use B.1(c).

**(c)** Let $U = \operatorname{span}\{[\,1,0\,]^{\mathsf{T}}\}$ and $W = \operatorname{span}\{[\,1,2\,]^{\mathsf{T}}\}$, subspaces of $\mathbb{R}^2$. Use the theorem of §29.3 to show that $\mathbb{R}^2 = U \oplus W$, and split $\mathbf{x} = [\,4,6\,]^{\mathsf{T}}$ into its piece in $U$ and its piece in $W$. Find the $2 \times 2$ matrix $\mathbf{P}$ that sends every $\mathbf{x}$ to its piece in $U$, check that $\mathbf{P}^2 = \mathbf{P}$, and say whether $\mathbf{P}$ is an orthogonal projection matrix.

**(d)** Let $W = \{\mathbf{x} \in \mathbb{R}^3 : x_1 + x_2 + x_3 = 0\}$, a subspace of $\mathbb{R}^3$. Show that $\mathbb{R}^3 = \operatorname{span}\{[\,1,1,0\,]^{\mathsf{T}}\} \oplus W$, and split $[\,3,1,2\,]^{\mathsf{T}}$ into its two pieces. Then show that $\mathbb{R}^3$ is *not* $\operatorname{span}\{[\,1,-1,0\,]^{\mathsf{T}}\} \oplus W$.

## Part C. Determinants

*Matrix Unit, §22–§23 and §35–§39.* The parallelogram with sides the columns of $\mathbf{A} \in \mathbb{R}^{2\times 2}$ has area $|\det \mathbf{A}|$ (§23.4), and $\mathbf{A}$ multiplies the area of every region by $|\det \mathbf{A}|$ (§23.6). For $\mathbf{A} \in \mathbb{R}^{n\times n}$, the minor $M_{ij}$ is the determinant of $\mathbf{A}$ with row $i$ and column $j$ deleted, and the cofactor is $C_{ij} = (-1)^{i+j} M_{ij}$ (§35). Then $\det \mathbf{A} = \sum_j a_{1j} C_{1j}$, and expanding along any other row or column gives the same number (§36).
The laws of §38.1 are:

| | | | |
|---|---|---|---|
| **D1** | $\det \mathbf{I}_n = 1$ | **D6** | two equal rows give $\det \mathbf{A} = 0$ |
| **D2** | $\det \mathbf{A}^{\mathsf{T}} = \det \mathbf{A}$ | **D7** | det is linear in each row separately |
| **D3** | a zero row gives $\det \mathbf{A} = 0$ | **D8** | adding a multiple of one row to another leaves det unchanged |
| **D4** | scaling one row by $\alpha$ scales det by $\alpha$; so $\det(\alpha\mathbf{A}) = \alpha^n \det \mathbf{A}$ | **D9** | $\det(\mathbf{AB}) = \det \mathbf{A} \det \mathbf{B}$ |
| **D5** | swapping two rows changes the sign | **D10** | $\det \mathbf{A} \neq 0 \iff \operatorname{rank} \mathbf{A} = n \iff \operatorname{null}[\mathbf{A}] = \{\mathbf{0}\}$ |

By D2, every law about rows is also a law about columns. The determinant of a triangular matrix is the product of its diagonal entries (§39.1). Keeping the rows *and* the columns numbered by a set $S \subseteq \{1,\dots,n\}$ gives a principal submatrix, whose determinant is a **principal minor**; for $S = \{1,\dots,k\}$ it is the $k$-th **leading principal minor** $\Delta_k$ (§37.1). A symmetric $\mathbf{A}$ is **positive definite** if $\mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{x} > 0$ for every $\mathbf{x} \neq \mathbf{0}$, which happens exactly when $\Delta_1, \dots, \Delta_n$ are all positive (Sylvester's criterion, §37.3).

**Problem C.1** (Area)**.**

**(a)** Find the area of the parallelogram with sides $[\,4,1\,]^{\mathsf{T}}$ and $[\,2,5\,]^{\mathsf{T}}$, and of the triangle with corners $(1,1)$, $(5,2)$ and $(3,6)$.

**(b)** Find every number $k$ for which the triangle with corners $(0,0)$, $(3,1)$ and $(k,2)$ has area 4.

**(c)** A region of the plane has area 5. Let $\mathbf{M} = \left[\begin{smallmatrix} 3 & 1 \\ 2 & 4 \end{smallmatrix}\right]$. Find the area of its image under $\mathbf{M}$, under $\mathbf{M}^2$, under $\mathbf{R}_{\pi/3}\mathbf{M}$, and under $\frac{1}{2}\mathbf{M}$, naming the facts of §22 and §23.6 you use.

**Problem C.2** (Computing determinants)**.**
Let

$$\mathbf{B} = \begin{bmatrix} 2 & -1 & 3 \\ 1 & 4 & 0 \\ -2 & 1 & 5 \end{bmatrix}, \qquad \mathbf{A} = \begin{bmatrix} 1 & 0 & 2 & 0 \\ 3 & 1 & 0 & 2 \\ 0 & 0 & 4 & 0 \\ 2 & 0 & 1 & 3 \end{bmatrix}, \qquad \mathbf{T}_k = \begin{bmatrix} 1 & k & 0 \\ k & 1 & k \\ 0 & k & 1 \end{bmatrix}.$$

**(a)** Compute $\det \mathbf{B}$ by expanding along the first row, and again by expanding down the third column.

**(b)** Compute $\det \mathbf{A}$, expanding along whichever row or column makes the work shortest (§36.4).

**(c)** Compute $\det \mathbf{B}$ a third time: reduce $\mathbf{B}$ to an upper triangular matrix by row operations, record the effect of each one on the determinant (§38.3), and finish with §39.1.

**(d)** Find every number $k$ for which $\det \mathbf{T}_k = 0$. For the positive one, find a nonzero vector $\mathbf{x}$ with $\mathbf{T}_k\mathbf{x} = \mathbf{0}$.

**Problem C.3** (Laws, two proofs, and positive definite matrices)**.**

**(a)** Let $\mathbf{A}$ be $3 \times 3$ with $\det \mathbf{A} = 4$. Find $\det(2\mathbf{A})$, $\det(-\mathbf{A})$, $\det(\mathbf{A}^3)$ and $\det(\mathbf{A}^{\mathsf{T}}\mathbf{A})$. Then find the determinant of the matrix obtained from $\mathbf{A}$ by (i) swapping rows 1 and 3; (ii) adding 5 times row 2 to row 1; (iii) tripling row 2 and then swapping rows 1 and 2. Name the law used each time.

**(b)** A square matrix $\mathbf{K}$ is **skew-symmetric** if $\mathbf{K}^{\mathsf{T}} = -\mathbf{K}$. Prove that if $\mathbf{K}$ is $n \times n$ and skew-symmetric and $n$ is odd, then $\det \mathbf{K} = 0$. Check it on $\left[\begin{smallmatrix} 0 & 1 & -2 \\ -1 & 0 & 3 \\ 2 & -3 & 0 \end{smallmatrix}\right]$, and show that the skew-symmetric $\left[\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right]$ has nonzero determinant.

**(c)** Prove that

$$\det \begin{bmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{bmatrix} = (b-a)(c-a)(c-b),$$

and deduce that this determinant is 0 exactly when two of $a, b, c$ are equal.

**Hint.** Subtract column 1 from columns 2 and 3 (D8 for columns), expand along the first row, and factor.

**(d)** Let $\mathbf{S} = \left[\begin{smallmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{smallmatrix}\right]$. Find all seven principal minors of $\mathbf{S}$ and its three leading principal minors. Is $\mathbf{S}$ positive definite?

**(e)** For $\mathbf{A} = \left[\begin{smallmatrix} 1 & 3 \\ 3 & 4 \end{smallmatrix}\right]$, compute $\Delta_1$ and $\Delta_2$, and find a vector $\mathbf{x}$ with $\mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{x} < 0$.

**Hint.** Look at the vector used in the proof of Sylvester's criterion in §37.3.

## Part D. The null space and rank–nullity

*Matrix Unit, §30–§34.* The **null space** of $\mathbf{A} \in \mathbb{R}^{m\times n}$ is $\operatorname{null}[\mathbf{A}] = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{A}\mathbf{x} = \mathbf{0}\}$, a subspace of $\mathbb{R}^n$, and its dimension is the **nullity** (§31). The matrix has **full column rank** if $\operatorname{rank} \mathbf{A} = n$, **full row rank** if $\operatorname{rank} \mathbf{A} = m$, and **full rank** if $\operatorname{rank} \mathbf{A} = \min(m, n)$ (§30.1). The rank–nullity theorem says $\operatorname{rank} \mathbf{A} + \operatorname{nullity} \mathbf{A} = n$ (§32). If $\mathbf{x}_p$ is one solution of $\mathbf{A}\mathbf{x} = \mathbf{b}$, the solutions are exactly $\mathbf{x}_p + \mathbf{z}$ with $\mathbf{z} \in \operatorname{null}[\mathbf{A}]$ (§31.6). If $\mathbf{r}$ is a row of $\mathbf{A}$ and $\mathbf{z} \in \operatorname{null}[\mathbf{A}]$, then $\mathbf{r}\mathbf{z} = 0$, so $\mathbf{z}$ is orthogonal to the column $\mathbf{r}^{\mathsf{T}}$ (§31.7), and $\operatorname{rank}(\mathbf{A}^{\mathsf{T}}\mathbf{A}) = \operatorname{rank} \mathbf{A} = \operatorname{rank} \mathbf{A}^{\mathsf{T}}$ (§33). For square $\mathbf{A}$, a subspace $W$ is **invariant** under $\mathbf{A}$ if $\mathbf{A}\mathbf{w} \in W$ for every $\mathbf{w} \in W$ (§30.4). If $ad - bc \neq 0$, then $\left[\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right]^{-1} = \frac{1}{ad-bc}\left[\begin{smallmatrix} d & -b \\ -c & a \end{smallmatrix}\right]$, and if $ad - bc = 0$ there is no inverse (§34).

**Problem D.1** (Null spaces, rank and nullity)**.**
Let

$$\mathbf{A}_1 = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix}, \quad \mathbf{A}_2 = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \\ 1 & 1 & 1 \end{bmatrix}, \quad \mathbf{A}_3 = \begin{bmatrix} 1 & 1 \\ 1 & -1 \\ 2 & 0 \end{bmatrix}, \quad \mathbf{A}_4 = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 3 \end{bmatrix}.$$

**(a)** For each $\mathbf{A}_i$, write $\operatorname{null}[\mathbf{A}_i]$ as the span of as few vectors as possible, or show that it is $\{\mathbf{0}\}$, and give the nullity.

**(b)** Give the rank of each $\mathbf{A}_i$ and check rank–nullity. Say whether each has full rank, full column rank, full row rank, or none of these.

**(c)** Check that $\mathbf{r}\mathbf{z} = 0$ for every row $\mathbf{r}$ of $\mathbf{A}_2$ and every vector $\mathbf{z}$ you used to span $\operatorname{null}[\mathbf{A}_2]$.

**(d)** Find every solution of $\mathbf{A}_2\mathbf{x} = [\,3, 1, 4\,]^{\mathsf{T}}$ in the form $\mathbf{x}_p + \mathbf{z}$, $\mathbf{z} \in \operatorname{null}[\mathbf{A}_2]$. Then show that $\mathbf{A}_2\mathbf{x} = [\,1, 1, 1\,]^{\mathsf{T}}$ has no solution.

**(e)** A matrix $\mathbf{A} \in \mathbb{R}^{6\times 9}$ has rank 4. Find $\operatorname{nullity}\mathbf{A}$, $\operatorname{rank}\mathbf{A}^{\mathsf{T}}$, $\operatorname{nullity}\mathbf{A}^{\mathsf{T}}$, $\operatorname{rank}(\mathbf{A}^{\mathsf{T}}\mathbf{A})$ and $\operatorname{rank}(\mathbf{A}\mathbf{A}^{\mathsf{T}})$, naming the result you use each time.

**Problem D.2** (Proofs, invariant lines, and the $2 \times 2$ inverse)**.**

**(a)** Let $\mathbf{A} \in \mathbb{R}^{m\times n}$ and $\mathbf{B} \in \mathbb{R}^{p\times m}$. Prove that $\operatorname{null}[\mathbf{A}] \subseteq \operatorname{null}[\mathbf{B}\mathbf{A}]$. Then prove that if $\operatorname{null}[\mathbf{B}] = \{\mathbf{0}\}$, then $\operatorname{null}[\mathbf{B}\mathbf{A}] = \operatorname{null}[\mathbf{A}]$.

**(b)** Let $\mathbf{A} \in \mathbb{R}^{m\times n}$ and $\mathbf{B} \in \mathbb{R}^{n\times p}$. Prove that $\operatorname{col}[\mathbf{A}\mathbf{B}] \subseteq \operatorname{col}[\mathbf{A}]$, and deduce that $\operatorname{rank}(\mathbf{A}\mathbf{B}) \le \operatorname{rank}\mathbf{A}$. Then use the first statement of (a), applied to the product $\mathbf{A}\mathbf{B}$, together with rank–nullity, to show that $\operatorname{rank}(\mathbf{A}\mathbf{B}) \le \operatorname{rank}\mathbf{B}$.

**(c)** Let $\mathbf{A}$ be $n \times n$. Prove that $\operatorname{null}[\mathbf{A}]$ and $\operatorname{col}[\mathbf{A}]$ are invariant under $\mathbf{A}$. Then find every line through the origin that is invariant under $\left[\begin{smallmatrix} 2 & 1 \\ 0 & 3 \end{smallmatrix}\right]$.

**(d)** For which numbers $k$ is $\mathbf{B}_k = \left[\begin{smallmatrix} k & 4 \\ 1 & k \end{smallmatrix}\right]$ invertible? For $k = 3$, find $\mathbf{B}_3^{-1}$ and use it to solve $\mathbf{B}_3\mathbf{x} = [\,1, 2\,]^{\mathsf{T}}$. For $k = 2$, find $\operatorname{null}[\mathbf{B}_2]$ and $\operatorname{col}[\mathbf{B}_2]$.

## Part E. Elimination and the inverse

*Matrix Unit, §40–§45 and §50–§53.* The elementary row operations are **(E1)** swap two rows, **(E2)** scale a row by a nonzero number, **(E3)** add a multiple of one row to another. They do not change the solutions of a system (§40.3), and each one is multiplication on the left by the **elementary matrix** obtained by doing it to $\mathbf{I}$ (§40.4). Row-echelon form is defined in §42.5 and reduced row-echelon form (RREF) in §43.1. By §43.4, the columns of $\mathbf{A}$ in the pivot positions of its RREF form a basis of $\operatorname{col}[\mathbf{A}]$, and setting one free variable to 1 and the others to 0, for each free variable in turn, gives a basis of $\operatorname{null}[\mathbf{A}]$. A system is consistent exactly when elimination produces no row $[\,0 \;\cdots\; 0 \mid c\,]$ with $c \neq 0$ (§44.3).

A matrix $\mathbf{A} \in \mathbb{R}^{n\times n}$ is **invertible** if there is a $\mathbf{B}$ with $\mathbf{A}\mathbf{B} = \mathbf{B}\mathbf{A} = \mathbf{I}$ (§51.1). Gauss–Jordan elimination turns $[\,\mathbf{A} \mid \mathbf{I}\,]$ into $[\,\mathbf{I} \mid \mathbf{A}^{-1}\,]$ (§51.4). The **adjugate** $\operatorname{adj}\mathbf{A}$ is the transpose of the matrix of cofactors, and $\mathbf{A}^{-1} = \frac{1}{\det\mathbf{A}}\operatorname{adj}\mathbf{A}$ (§52). If $\det\mathbf{A} \neq 0$, **Cramer's rule** gives the solution of $\mathbf{A}\mathbf{x} = \mathbf{b}$ as $x_i = \det\mathbf{A}_i(\mathbf{b}) / \det\mathbf{A}$, where $\mathbf{A}_i(\mathbf{b})$ is $\mathbf{A}$ with column $i$ replaced by $\mathbf{b}$ (§50.1). The laws of §53.1 are **I1** $(\mathbf{A}^{-1})^{-1} = \mathbf{A}$, **I2** $(\mathbf{A}\mathbf{B})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$, **I3** $(\mathbf{A}^{\mathsf{T}})^{-1} = (\mathbf{A}^{-1})^{\mathsf{T}}$, **I4** $(\alpha\mathbf{A})^{-1} = \alpha^{-1}\mathbf{A}^{-1}$, **I5** $\det(\mathbf{A}^{-1}) = 1/\det\mathbf{A}$ and **I6** $(\mathbf{A}^{k})^{-1} = (\mathbf{A}^{-1})^{k}$.

**Problem E.1** (Elimination, step by step)**.**

**(a)** Solve the system

$$x_1 + 2x_2 + x_3 = 1, \qquad 2x_1 + 5x_2 + 3x_3 = 3, \qquad -x_1 + x_2 + 4x_3 = 6$$

by Gaussian elimination and back substitution, writing down each row operation. Then read off the determinant of the coefficient matrix $\mathbf{A}$ from the pivots (§42.6).

**(b)** Write the elementary matrix of each row operation you used in (a). Check that their product, in the right order, times $\mathbf{A}$ is the upper triangular matrix you reached.

**(c)** Solve $\mathbf{L}\mathbf{x} = [\,4, 5, 3\,]^{\mathsf{T}}$ by forward substitution (§41.2), where $\mathbf{L} = \left[\begin{smallmatrix} 2 & 0 & 0 \\ 1 & 3 & 0 \\ -1 & 2 & 1 \end{smallmatrix}\right]$.

**Problem E.2** (Reduced row-echelon form, bases, and consistency)**.**
Let

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 0 & 1 & -1 \\ 2 & 4 & 1 & 4 & 1 \\ 1 & 2 & 1 & 3 & 2 \end{bmatrix}.$$

**(a)** Bring $\mathbf{A}$ to reduced row-echelon form, naming each row operation. Mark the pivots, and list the pivot variables and the free variables.

**(b)** Use §43.4 to give a basis of col[$\mathbf{A}$] made of columns of $\mathbf{A}$, and a basis of null[$\mathbf{A}$]. Give the rank and the nullity, and check rank–nullity.

**(c)** Write each non-pivot column of $\mathbf{A}$ as a combination of the pivot columns of $\mathbf{A}$, and check one of them by computing.

**(d)** Find a condition on $b_1, b_2, b_3$ that holds exactly when $\mathbf{A}\mathbf{x} = [\,b_1, b_2, b_3\,]^{\mathsf{T}}$ is consistent. Then find every solution of $\mathbf{A}\mathbf{x} = [\,2, 5, 3\,]^{\mathsf{T}}$, and show that $\mathbf{A}\mathbf{x} = [\,1, 1, 1\,]^{\mathsf{T}}$ has none.

**Problem E.3** (The inverse)**.**
Let

$$\mathbf{M} = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \\ 2 & 4 & 1 \end{bmatrix}, \qquad \mathbf{N} = \begin{bmatrix} 2 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 3 & 1 \end{bmatrix}.$$

**(a)** Find $\mathbf{M}^{-1}$ by Gauss–Jordan elimination, and check that $\mathbf{M}\mathbf{M}^{-1} = \mathbf{I}$.

**(b)** Find $\mathbf{N}^{-1}$ by the adjugate, and check that $\mathbf{N}\mathbf{N}^{-1} = \mathbf{I}$.

**(c)** Use Cramer's rule to find $x_2$ alone in $\mathbf{N}\mathbf{x} = [\,1, 3, 5\,]^{\mathsf{T}}$. Then confirm it with $\mathbf{N}^{-1}$ from (b).

**(d)** Let $\mathbf{A}, \mathbf{B}, \mathbf{C}$ be invertible $n \times n$ matrices. Prove that $(\mathbf{A}\mathbf{B}\mathbf{C})^{-1} = \mathbf{C}^{-1}\mathbf{B}^{-1}\mathbf{A}^{-1}$, naming the laws of §8.4 and §53.1 you use. Then prove that if $\mathbf{A}$ is invertible and $\mathbf{A}\mathbf{B} = \mathbf{A}\mathbf{C}$, then $\mathbf{B} = \mathbf{C}$.

**(e)** Let $\mathbf{A}$ be $n \times n$ with $\mathbf{A}^2 = \mathbf{0}$. Prove that $\mathbf{A}$ is not invertible, and that $\mathbf{I} - \mathbf{A}$ is invertible with $(\mathbf{I} - \mathbf{A})^{-1} = \mathbf{I} + \mathbf{A}$. Check both statements on $\mathbf{A} = \left[\begin{smallmatrix} 1 & -1 \\ 1 & -1 \end{smallmatrix}\right]$.

## Part F. Least squares and the pseudo-inverse

*Matrix Unit, §46–§49 and §54.* A **least-squares solution** of $\mathbf{Ax} = \mathbf{b}$ is a vector $\hat{\mathbf{x}}$ that makes $\|\mathbf{Ax} - \mathbf{b}\|^2$ as small as possible (§47.1). These are exactly the solutions of the **normal equations** $\mathbf{A}^\mathsf{T}\mathbf{A}\,\hat{\mathbf{x}} = \mathbf{A}^\mathsf{T}\mathbf{b}$, which say that the residual $\mathbf{b} - \mathbf{A}\hat{\mathbf{x}}$ is orthogonal to every column of $\mathbf{A}$ (§47.2–§47.3). The normal equations always have a solution, and it is unique exactly when $\mathbf{A}$ has full column rank (§49.1–§49.2). To fit the line $y = c_0 + c_1 t$ to data points $(t_i, y_i)$, row $i$ of $\mathbf{A}$ is the row vector $[\,1, t_i\,]$ and $\mathbf{b}$ lists the $y_i$ (§49.4). The **pseudo-inverse** is $\mathbf{A}^+ = (\mathbf{A}^\mathsf{T}\mathbf{A})^{-1}\mathbf{A}^\mathsf{T}$ if $\mathbf{A}$ has independent columns (§54.2), and $\mathbf{A}^+ = \mathbf{A}^\mathsf{T}(\mathbf{A}\mathbf{A}^\mathsf{T})^{-1}$ if $\mathbf{A}$ has independent rows, in which case $\mathbf{A}^+\mathbf{b}$ is the shortest solution of $\mathbf{Ax} = \mathbf{b}$ (§54.3). For any $\mathbf{A}$, $\mathbf{A}^+$ is the one matrix $\mathbf{X}$ with **(MP1)** $\mathbf{AXA} = \mathbf{A}$, **(MP2)** $\mathbf{XAX} = \mathbf{X}$, **(MP3)** $(\mathbf{AX})^\mathsf{T} = \mathbf{AX}$ and **(MP4)** $(\mathbf{XA})^\mathsf{T} = \mathbf{XA}$ (§54.4).

**Problem F.1** (Fitting a line)**.**
The four data points are $(0, 1)$, $(1, 3)$, $(2, 4)$ and $(3, 4)$.

**(a)** Write the system $\mathbf{Ax} = \mathbf{b}$, with $\mathbf{x} = [\,c_0, c_1\,]^\mathsf{T}$, that says the line $y = c_0 + c_1 t$ passes through all four points. Show by elimination that it has no solution.

**(b)** Form $\mathbf{A}^\mathsf{T}\mathbf{A}$ and $\mathbf{A}^\mathsf{T}\mathbf{b}$, solve the normal equations, and write down the best line.

**(c)** Compute the residual $\mathbf{r} = \mathbf{b} - \mathbf{A}\hat{\mathbf{x}}$, check that $\mathbf{A}^\mathsf{T}\mathbf{r} = \mathbf{0}$, and find the sum of the squared errors. Sketch the four points, the line, and the four vertical segments whose lengths are the entries of $\mathbf{r}$.

**(d)** Compute $\mathbf{A}^+ = (\mathbf{A}^\mathsf{T}\mathbf{A})^{-1}\mathbf{A}^\mathsf{T}$. Check that $\mathbf{A}^+\mathbf{A} = \mathbf{I}_2$ and that $\mathbf{A}^+\mathbf{b} = \hat{\mathbf{x}}$.

**Problem F.2** (Pythagoras, shortest solutions, and Moore–Penrose)**.**

**(a)** Let $\hat{\mathbf{x}}$ be a least-squares solution of $\mathbf{Ax} = \mathbf{b}$. Prove that $\|\mathbf{b}\|^2 = \|\mathbf{A}\hat{\mathbf{x}}\|^2 + \|\mathbf{b} - \mathbf{A}\hat{\mathbf{x}}\|^2$. Check it on Problem F.1.

**(b)** Use §54.3 to find the shortest solution $\mathbf{x}^+$ of $x_1 + 2x_2 + 2x_3 = 9$. Check that $\mathbf{x}^+$ is orthogonal to every vector of the null space of the $1 \times 3$ matrix $[\,1, 2, 2\,]$, and compare its length with that of the solution $[\,9, 0, 0\,]^\mathsf{T}$.

**(c)** The matrix $\mathbf{A}$ below has neither independent columns nor independent rows:

$$\mathbf{A} = \begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix}, \qquad \mathbf{X} = \frac{1}{25}\begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix}.$$

Check that $\mathbf{X}$ satisfies MP1–MP4, so $\mathbf{X} = \mathbf{A}^+$. Show that $\mathbf{AX}$ and $\mathbf{XA}$ are the projection matrices $\mathbf{P_y}$ and $\mathbf{P_z}$ of §26.2 for $\mathbf{y} = [\,2, 1\,]^\mathsf{T}$ and $\mathbf{z} = [\,1, 2\,]^\mathsf{T}$.

**(d)** With $\mathbf{A}$ from (c) and $\mathbf{b} = [\,5, 0\,]^\mathsf{T}$, show that the normal equations have infinitely many solutions, and find them all. Show that $\mathbf{A}^+\mathbf{b}$ is one of them, and that it is shorter than the least-squares solution $[\,2, 0\,]^\mathsf{T}$.

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Questions about any problem here belong in the class Discord, where everyone can see the answer. Questions that are specific to your own work — "is my proof of D.2(b) valid?" — are better brought to office hours, Thursdays 3:00–4:00 PM in Kiely Hall 508. Please email ahead.